

CPS643 - User Manual

Jeremy Mathew - 501166347

Department of Computer Science, Toronto Metropolitan University

CPS643 02: Virtual Reality

Tim McInerney

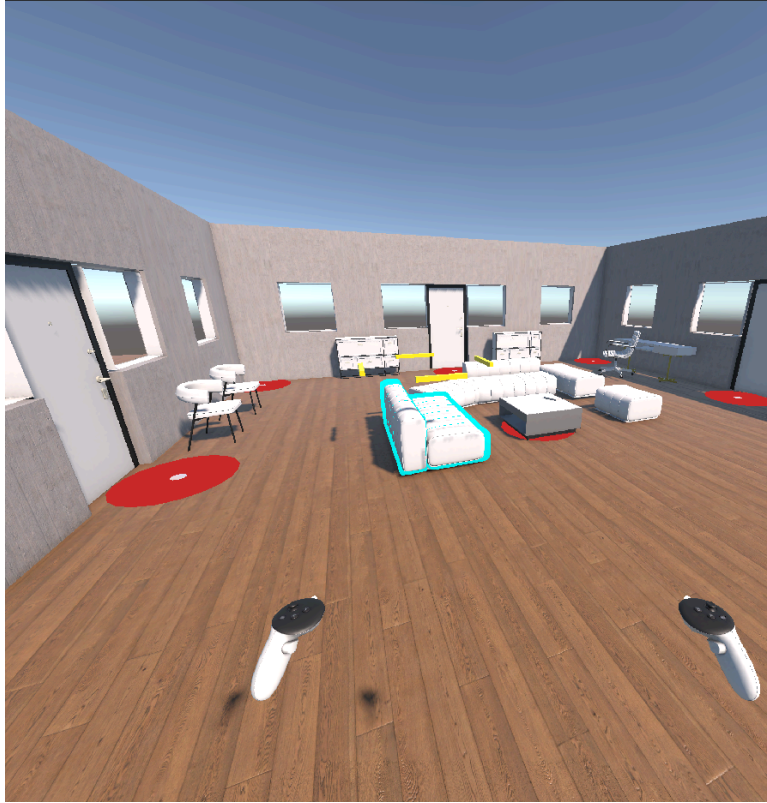
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Project Description

The system is a VR-based interior design software program enabling users to interact with and make changes to a 3D room environment in real-time. Through the use of VR controllers, users can pick up and manipulate objects, create and customize walls made up of modules, and alternate between modes of interaction such as moving objects, wall panels, and creating the environment. Navigation in the system includes teleportation hotspots and miniature world navigation, allowing users to better comprehend and navigate through the whole environment.

Controls:

- **Right Trigger (Ray Select):** Select and interact with furniture and wall panels using a laser pointer
- **Right Grip:** Grab and drag furniture or objects in the scene (Coarse or Fine mode)
- **Right Grip:** Move objects using grid-based stepping (Object Grid mode)
- **Right A Button:** teleport to hotspot
- **Right B Button:** Toggle VR widgets (movement, panel editing, creation tools)
- **Left Thumbstick Click:** Toggle World-in-Miniature (WIM) view
- **Head Movement:** Look direction controls/teleport hotspot selection and initial widget orientation



VR System for Room Interior Design

Interior design of a room using widgets (multi-scale) in VR for precise layout/dimensions of walls, doors, windows, and the use of WIM + widgets.